

Push 2 / Nuendo Bridge

Release Notes

Version 1.0.5 — May 2026

New Features

- **Keyswitch pad layouts:** the Layout button now cycles through four pad layouts — 64 notes, 56 notes + 8 keyswitches, 48 notes + 16 keyswitches, and Drum. Keyswitch pads sit on the bottom row(s), light orange, and trigger user-defined absolute notes (ideal for articulation keyswitches). Long-press Layout to configure: Enc 1 sets the starting note, Enc 2-8 override individual pads, and the lower row toggles Chromatic / Naturals layout, Latch (the keyswitch section is monophonic — only one active at a time), Reset, and Done. Note names follow Cubase convention (C-2 = MIDI 0). Configuration is in-memory (not persisted).
- **Two octatonic scales:** *Octatonic WH* and *Octatonic HW* added to the Scale menu.
- **XY pad** (Session button): the 64 pads become an XY morphing surface controlling two parameters of the selected track (Volume, Pan, Quick Controls 1-8) or raw MIDI CC. Relative, pressure-weighted input means no jump when you touch down, and two-finger input interpolates smoothly. Lower 1/2 pick the X/Y category (Track / CC), Enc 1/2 pick the X/Y parameter, Enc 4/5 set sensitivity and smoothing, Lower 5-8 toggle Mute/Solo/Monitor/Record for the selected track, and the arrows navigate tracks.
- **Pad note range in the Mix footer:** the Mix page footer now shows the MIDI note range currently mapped to the pads (keyswitch pads excluded in keyswitch layouts).
- **Per-track automation feedback:** each bank track's Read/Write automation mode is now reflected on screen, not just the selected track's.
- **Windows System Tray app:** on Windows the bridge now runs from the system tray (status, Open Plugin Mapper, Show Log, Quit) instead of a console window. Run with `--terminal` for the classic console version. The Plugin Mapper no longer opens a browser automatically.
- **Rescan moved to the Setup page** (lower-row button 8, available on every Setup page).
- **macOS app is now fully self-contained:** the libusb USB runtime is bundled inside the .app, so Homebrew and `brew install libusb` are no longer required — just copy the app and run it.

Bug Fixes

- **Control Room Main reset** (Shift+Touch on the rightmost encoder) now lands on an exact 0.00 dB instead of -0.01 dB.
- **Channel Strip module values** now appear immediately on entering a module (Gate, Comp, EQ, etc.) and stay visible, instead of only showing after you nudge a parameter.
- **Saturator and Limiter encoders** now respond to changes (the sub-page activation could fail silently for those two slots).
- **Inserts page:** an insert is no longer occasionally shown duplicated in another slot (display only).
- **CC Pick-up mode removed:** the MIDI CC page is Absolute only. Pick-up required parameter feedback that a normal setup does not provide, so it behaved identically to Absolute.

Version 1.0.4 — May 2026

New Features

- **Channel Strip mode** (Mix → Mix again): a dedicated Channel Strip view with an overview page (Gate, Comp, EQ, Tools, Sat, Limiter shown as Cubase-coloured banners) and six drill-down sub-pages — one per module.
- **Strip slot extended parameters**: parameters that are NOT in Cubase's bank zone are now reachable via DirectAccess on every strip variant — e.g. VintageCompressor Ratio & Mix, Standard Compressor DryMix & Hold, Tube Compressor Character & High Ratio, DeEsser side-chain filter & Diff, Magneto II Dual & Oversample, Maximizer Release & Recover, Brickwall Link & Oversample, Noise Gate Hold & Analysis, and more.
- **EQ page (EQ-Eight style)**: band selector (Enc 1), Type / Freq / Q / Gain for the selected band, PreFilter Low-Cut / High-Cut frequency and on/off, PreGain, plus a live magnitude curve (4 EQ bands + PreFilter) with numbered band markers. The curve updates in real time from the Push **and** from changes made in Nuendo.
- **Edit Channel Settings**: now on lower-row button 8 (Channel Strip pages) and Shift+Upper Row (Mix view), in addition to the existing double-press. The button lights white and the on-screen pill turns white while the window is open (bidirectional state feedback).
- **Section colours & bold labels**: each Channel Strip section uses its Nuendo track-strip colour with bold module names, matching the Cubase/Nuendo GUI.
- **Amber Bypass indicators**: bypass pills/LEDs light amber when the section is bypassed (engaged), matching Nuendo's bypass button behaviour.
- **Lower-row LED sync**: lower-row button LEDs follow the on-screen toggle states on the Channel Strip overview and all sub-pages.
- **Piano scale mode**: a new *Piano* option in the Scale selector lays the pads out like a piano keyboard — white keys on the lower row of each octave pair, black keys on the upper row, four octaves stacked bottom-to-top, every C highlighted in purple. Rooted at C (root selection is locked in this mode); Octave Up/Down shifts the whole four-octave block.

Technical

- Strip slot DA exploration (up to 6 slots) with dynamic DA-slot resolution by plugin name — supports custom Cubase strip layouts where module order differs.
- DA param-flip mechanism (CC 73/74 ch8) for binary toggles that Cubase's setTypeToggle handles unreliably (e.g. VintageCompressor Punch, EQ section bypass, EQ band on/off).
- Live host-side parameter feedback via SysEx 0x3D from daInserts.mOnParameterChange, de-duplicated to avoid flooding during mouse drags in Nuendo.
- Edit Channel Settings window state via SysEx 0x3E (separate read binding).
- Dedicated read bindings for PreFilter Phase / section bypass and ChannelEQ bypass so the on-screen state stays in sync.
- SysEx range 0x34–0x3E and ch8 CC 70–78 allocated for strip slot cache, bypass, edit, param-flip, variant probe and live feedback.

Known Limitations

- **Windows — the bridge and Ableton Live cannot use the Push 2 at the same time.** Windows installs the Push 2's WinUSB driver automatically (no Zadig needed); the bridge and Live share that same driver, but only one app can hold the USB connection at a moment — close one to use the other. (macOS is not affected.)
- Channel Strip modules must be **activated manually in Nuendo** first — an empty or disabled strip slot (Gate, Comp, EQ, Tools, Sat, Limiter) is not exposed to the Push 2. Load/enable the module in the Nuendo channel strip before controlling it from the controller.
- Strip slot **variant switching from the Push is not possible** — Cubase locks strip-slot plugin selection; the MIDI Remote API (trySetSlotPlugin / the slot 'Effect Type' tag) is silently rejected. Switch variants in Nuendo's strip-slot menu; the on-screen chip mirrors the active variant in real time.
- Opening a strip-slot plugin's **Edit UI from the Push is not exposed** by the MIDI Remote API (mEdit toggles a flag but Nuendo does not open the window). Use the 'e' button in the mixer console.
- ChannelEQ **bands 2 and 3 are always Peak** filters (Cubase restriction) — the Type encoder has no effect on those bands.
- No audio spectrogram behind the EQ curve — the MIDI Remote API provides no audio stream.
- **Remove the Push 2 'Live Port' and 'User Port' from Nuendo's "In All MIDI Inputs"** (Studio Setup → MIDI Port Setup) or pad notes will be doubled (one copy from the bridge's BridgeNotes port, one direct from the hardware).

Version 1.0.3 — April 2026

New Features

- **Plugin Browser** (Add Device button): browse and load plugins directly from the Push 2. Select an insert slot, scroll through your plugin collection, and load with one press. Uses the DirectAccess Plugin Manager API (Nuendo 15+ / API 1.3).
- **Collection cycling**: switch between Nuendo plugin collections via lower row button 1 in the plugin list view.
- **Plugin Mapper integration**: the Plugin Mapper web server (FastAPI) is now integrated into the bridge and starts automatically. Access it from the menu bar or at <http://localhost:8100>. Dependencies are optional.
- **DirectAccess insert control**: bypass, edit (open/close UI), and encoder parameter control via DirectAccess for instant response. Fallback to viewer-based control if DA is not available.
- **DA encoder control**: mapped plugin parameters controlled directly via DirectAccess on MIDI channel 9, with display value feedback.
- **Plugin Mapper fuzzy matching**: pedalboard parameter names matched to DA parameter names using word-level fuzzy matching.
- **Record button LED**: 3-state behavior — orange (idle), blinking red (track armed), solid red (recording).
- **Shift+Arrows nudge**: in Volume and Pan modes, Shift+Left/Right shifts the bank by 1 track instead of 8.
- **CC Mode pick-up** (Setup page): new CC Mode page with Absolute and Pick-up options. Pick-up mode prevents parameter jumps by waiting for a direction change before sending.
- **Adaptive text color**: track headers and title bars automatically switch between white and black text based on background luminance. Bold font for headers.
- **Splash screen updated**: removed IAC port reference. Now shows 'Load the MIDI Remote script in Nuendo'.

Technical

- Plugin list transfer via SysEx 0x2C/0x2D, queued in mOnIdle (10 entries/tick).
- Plugin load via CC 80-82 ch8 + trySetSlotPlugin (DA Plugin Manager API 1.3).
- Record state feedback via CC 73.
- Bank shift via CC 38/39 bound to mShiftRight/mShiftLeft.
- Plugin Mapper server in daemon thread with dedicated asyncio event loop.
- MIDI channels 8 (0xB7) and 9 (0xB8) allocated for DirectAccess.

Known Limitations

- No Effect (remove plugin) not supported — API 1.3 has no removeSlotPlugin.
- CC pick-up uses direction-change detection (no Nuendo feedback available).
- Plugin Browser requires Nuendo 15+ (API 1.3).

Version 1.0.2 — April 2026

- Setup Page with Aftertouch Mode and Velocity Curve settings.
- MIDI CC Controller page (Shift+Note): 8 assignable CC faders.
- Mute/Solo/Monitor/Rec on Device and Inserts pages.
- Long press / double press on upper row buttons.
- Clear Monitor / Clear Rec (Shift+Mute/Solo).

Version 1.0.1 — March 2026

- Volume stutter fix, peak clip grace period, Send 1 CC conflict fix, CR CC conflict fix, pan offset correction.

Version 1.0.0 — February 2026

- Initial release: Mixer, Sends, Device, Inserts, Transport, Automation, Control Room, Note Input, Touchstrip, Track Management, VU Meters, macOS Menu Bar App.